

COLLECTIVE AI | PORTFOLIO THEORY | GOVERNED AUTONOMY

FROM DISTRIBUTED JUDGMENT TO A GOVERNED FINANCIAL PORTFOLIO

A 50-Agent Swarm Architecture for Financial Research, Collective Conviction, and Constrained Allocation

A pedagogical and formal companion to the executable notebook

NB_50_AGENT_FINANCIAL_SWARM_CLAUDE_SONNET_5.ipynb¹. The paper develops the financial logic of a system in which fifty Claude Sonnet 5 agents generate heterogeneous judgments over a common market state, while deterministic code validates, aggregates, measures disagreement, estimates covariance, and constructs the final long-only portfolio under explicit constraints.

PAPER PURPOSE

The central question is not whether a language model can “pick stocks.” It is whether a large collection of bounded, heterogeneous AI agents can be organized as a research institution whose judgments are useful inputs to a conventional portfolio process. The companion notebook makes this separation explicit. Market data and feature engineering are deterministic; fifty agents receive the same evidence but different mandates; native structured outputs impose a machine-readable contract; robust aggregation converts opinions into a conviction statistic; and a constrained mean–variance program converts that conviction into portfolio weights. The resulting architecture is therefore best understood as a hybrid of collective intelligence and classical quantitative finance.

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September 2026

AI for Financial Practitioners | Quantitative Finance

Research Paper

Companion computational artifact: NB_50_AGENT_FINANCIAL_SWARM_CLAUDE_SONNET_5.ipynb. Educational and research use; not investment advice and not a production investment-management specification.

¹The companion notebook is available at <https://drive.google.com/file/d/1kJK8-8Vnkz2GjswP7rVntu1tGMYJlhvv/view?usp=sharing>.

Abstract

This paper presents the methodological foundations of a 50-agent large-language-model swarm implemented in the companion notebook `NB_50_AGENT_FINANCIAL_SWARM_CLAUDE_SONNET_5.ipynb`. It examines how heterogeneous artificial-intelligence agents can transform a common panel of historical equity evidence into research judgments while deterministic methods retain authority over validation, risk measurement, constraints, and portfolio allocation. The universe contains thirty liquid U.S. equities. Three years of adjusted daily prices are converted into transparent features, including trailing returns, annualized volatility, downside volatility, maximum drawdown, risk-adjusted momentum, correlations, and an annualized covariance matrix. Agents receive this shared numerical state; they neither retrieve independent market facts nor control execution.

The swarm comprises ten specialist families with five variants each: momentum, reversal, defensive, drawdown, efficiency, diversification, cross-sectional, regime, skeptical, and balanced. Heterogeneity therefore arises from institutional mandates rather than arbitrary sampling noise. Each agent returns a validated structured assessment of every security, including a score, confidence measure, rationale, portfolio thesis, and portfolio-level risks. These outputs form an auditable opinion tape. Collective conviction combines the median score and average confidence, then penalizes cross-agent dispersion. The resulting statistic rewards directional agreement while treating dissent as economically meaningful uncertainty.

Portfolio construction remains outside the generative layer. Securities with positive conviction enter a candidate set; normalized conviction becomes an alpha-like preference signal; historical covariance supplies the risk model; and deterministic SLSQP optimization maximizes linear reward less quadratic variance subject to full investment, long-only holdings, and a 15% position cap. A convergence audit compares smaller random subswarms with the full-swarm ranking using Spearman correlation to assess whether additional agents stabilize collective ordering.

The paper’s contribution is architectural rather than predictive. It demonstrates how generative judgment can operate upstream of classical portfolio theory within explicit evidence, interface, allocation, and audit boundaries. The notebook is not an investable-strategy backtest, proof of return forecasting, or evidence that self-reported confidence is calibrated. Its retrospective market packet, heuristic conviction rule, and internal convergence benchmark preclude such claims. Instead, the system provides an inspectable foundation for prospective research using time-stamped data vintages, rolling out-of-sample decisions, realistic costs, benchmark portfolios, confidence calibration, alternative models, and tests of whether disagreement predicts subsequent uncertainty or returns.

CORE ARCHITECTURAL CLAIM

The notebook does not delegate the investment process to fifty autonomous stock pickers. It decomposes the process into distinct authorities: data establishes the common state; specialist agents supply heterogeneous judgments; validation controls admissibility; robust aggregation produces a collective preference; covariance and constraints determine feasible capital allocation; and audit measures whether the collective ranking stabilizes as the swarm grows.

The Research Problem: Can Collective AI Be Made Financially Governable?

Portfolio management is already a collective-intelligence problem. A sophisticated investment institution distributes work among researchers, sector specialists, macro strategists, risk managers, portfolio managers, traders, compliance officers, and investment committees. These participants rarely possess identical information or identical loss functions. Their institutional value comes partly from specialization and partly from the controlled reconciliation of disagreement. The 50-agent notebook translates this organizational fact into software.

The research question is consequently more precise than asking whether AI can forecast returns. It is:

Can a large but governed population of heterogeneous language-model agents transform a common, auditable market state into a useful cross-sectional preference signal, while deterministic portfolio mathematics retains authority over risk, constraints, and final weights?

This framing matters because financial decisions contain qualitatively different computational tasks. Measuring a twelve-month return is not a judgment problem. Computing an annualized covariance matrix is not a rhetorical task. Enforcing $\sum_i w_i = 1$ or $0 \leq w_i \leq 0.15$ should not depend on a model's linguistic compliance. By contrast, deciding whether a return pattern looks robust across horizons, whether drawdown risk should dominate recent momentum, or whether a ranking appears fragile to regime change is closer to the domain of contextual judgment. The notebook therefore assigns deterministic and generative components different institutional rights.

The design can be represented schematically as

$$\mathcal{D} \longrightarrow \mathcal{X} \longrightarrow \{A_j(\mathcal{X})\}_{j=1}^{50} \longrightarrow \mathcal{V} \longrightarrow \mathcal{G} \longrightarrow \mathcal{P}, \quad (1)$$

where $\mathcal{D}$ is market data, $\mathcal{X}$ the engineered state, A_j agent j , $\mathcal{V}$ the validation layer, $\mathcal{G}$ the collective aggregation rule, and $\mathcal{P}$ the deterministic portfolio program. No arrow can be skipped without changing the epistemic meaning of the system.

The phrase “financially governable” refers to more than numerical feasibility. It means that a reviewer can determine which component observed which information, which transformations were deterministic, where subjective judgment entered, and which mechanism had final authority over capital. In the notebook, governance begins with a deliberately narrow evidence packet. All agents receive the same feature table and correlation information, so an opinion can be traced to a common factual substrate. Their mandates differ, but their access rights do not. This arrangement makes disagreement interpretable: a divergence between a momentum specialist and a drawdown specialist reflects different objectives applied to shared evidence rather than an unknown difference in data access.

Governability also requires that every layer be capable of refusing an invalid input. The data layer can exclude securities with insufficient history. The validation layer can reject malformed or truncated model responses. The aggregation layer can count contributing agents and reveal missing coverage. The optimization layer can report failure rather than silently manufacturing weights. These rejection mechanisms matter because complex systems often fail through plausible-looking intermediate objects. A fluent rationale with an invalid ticker, an out-of-range score, or an incomplete security list should not acquire authority merely because it is readable. The system treats language-model output as proposed data, not trusted data.

The institutional analogy helps distinguish distributed intelligence from distributed authority. Research organizations benefit when specialists examine a decision from several perspectives, but they do not normally permit every analyst to alter positions directly. Recommendations pass through review, portfolio construction, trading controls, and compliance. The notebook recreates that separation in simplified form. Agents may express preferences and risks; code determines whether those expressions satisfy the contract; aggregation summarizes the committee; and the optimizer alone maps an eligible signal into weights. This design supports experimentation because one layer can be modified without implicitly changing all others. A researcher can replace the conviction function, covariance estimator, or agent taxonomy and then identify which intervention changed the result.

Finally, the research problem contains two distinct standards of success. An internal standard asks whether the architecture is coherent, reproducible, robust to malformed outputs, and stable as agents are added. An external standard asks whether its decisions improve realized risk-adjusted performance after costs. The present artifact primarily addresses the internal standard. That is a necessary first step: a system whose provenance and authority cannot be reconstructed should not be evaluated as though a favorable return were sufficient evidence. Economic validation comes later, under frozen protocols and out-of-sample data.

This framing also establishes a burden of proof appropriate to finance. New intelligence should not receive new authority merely because its outputs are articulate. It should first demonstrate controlled behavior, then incremental information, and finally economic value under realistic implementation. The notebook addresses the first stage while creating the records required for the next two.

Classical Portfolio Theory as the Deterministic Backbone

The notebook is modern in its research interface but classical in its capital-allocation logic. Markowitz’s central insight was that portfolio choice depends on joint distributions rather than stand-alone security characteristics [9]. If $w \in \mathbb{R}^n$ denotes portfolio weights and Σ the covariance matrix of returns, portfolio variance is

$$\sigma_p^2 = w^\top \Sigma w. \quad (2)$$

This quadratic form is the mathematical reason diversification cannot be reduced to choosing the individually “best” assets. Two securities with similar expected rewards can have radically different portfolio value because their covariance contributions differ.

The notebook preserves this principle. The swarm’s conviction statistic does not directly become a weight. It becomes an alpha-like preference vector α . The portfolio layer then solves

$$\max_w \quad \alpha^\top w - \lambda w^\top \Sigma w, \quad (3)$$

subject to

$$\mathbf{1}^\top w = 1, \quad 0 \leq w_i \leq \bar{w}, \quad (4)$$

where the notebook uses $\lambda = 4$ and $\bar{w} = 0.15$. Equation (3) is not presented as an optimal model of investor utility. It is a transparent teaching device that forces the swarm’s preferences to compete with measured covariance risk.

This separation also responds to a familiar empirical problem: estimated expected returns are extremely noisy. Mean–variance weights can be unstable when small changes in inputs produce large changes in allocations [10]. The notebook does not solve this general problem, but it limits one source of instability by min–max normalizing collective conviction and imposing long-only and concentration bounds. In production, one would consider shrinkage covariance estimators [8], turnover penalties, transaction costs, factor exposures, benchmark-relative risk, liquidity constraints, and perhaps Bayesian or Black–Litterman-style blending [2].

The role of portfolio theory is especially important because a cross-sectional ranking and a portfolio are different mathematical objects. A ranking answers which securities appear more attractive under a chosen signal. A portfolio must additionally decide how much capital to assign, how correlated positions interact, and which allocations are permitted. If the highest-conviction securities are concentrated in one economic theme, weighting them solely by conviction can create a portfolio whose apparent diversity is cosmetic. The quadratic risk term penalizes combinations that move together and therefore forces the allocation layer to consider joint exposure rather than isolated scores.

The risk-aversion coefficient λ provides an explicit policy lever. A smaller value permits the reward vector to dominate, producing a portfolio more concentrated in high-conviction names until position caps bind. A larger value places greater emphasis on covariance risk and generally encourages more balanced allocations. Neither setting is objectively correct without an investor mandate. The important point is that the trade-off is stated numerically and can be subjected to sensitivity analysis. A professional extension would report an efficient frontier across several values of λ , compare ex ante volatility, and document how frequently each constraint becomes active.

Constraints should likewise be interpreted as part of the investment thesis rather than incidental implementation details. Long-only status rules out negative weights and the leverage they can imply. Full investment prevents the optimizer from avoiding uncertainty by holding unallocated cash. The 15% cap ensures that at least seven positions are required for a feasible fully invested solution, although the realized number can be larger. These rules encode governance choices about concentration and mandate. Changing them can alter the portfolio more than modest changes in conviction, which is why constraint configurations belong in the reproducibility record.

Historical covariance introduces its own estimation risk. Daily observations may overweight the most recent regime, correlations can rise during stress, and a sample covariance matrix may be ill-conditioned when the universe is large relative to the observation window. The notebook’s three-year history and modest candidate set make the demonstration tractable, but they do not eliminate regime dependence. Shrinkage toward a structured target, factor-based covariance, stress matrices, or robust estimators would provide useful comparisons. The optimizer should also be tested under perturbed covariance inputs to determine whether seemingly precise weights are stable.

Thus the deterministic backbone does not claim to remove uncertainty. It converts uncertainty into explicit modeling choices: the reward proxy, the covariance estimate, the risk penalty, and the constraint set. This transparency is preferable to asking a generative model to reconcile all objectives implicitly in prose. Classical portfolio theory supplies the accounting system through which heterogeneous research judgments become a feasible, reviewable allocation.

For this reason, the optimizer should be viewed as a policy implementation engine rather than an oracle. It returns the best solution to the objective and inputs supplied; it cannot determine whether those inputs deserve belief. Its precision is computational, not epistemic. Keeping that distinction visible prevents exact weights from conveying more certainty than the research process supports.

Code Units 1–3: From Market Data to a Common Analytical State

Environment, secrets, and reproducibility

Code Unit 1 installs the Anthropic and market-data libraries, retrieves `ANTHROPIC_API_KEY` from Colab Secrets, defines `claude-sonnet-5`, fixes a numerical random seed, and declares a thirty-security U.S. equity universe. The API credential is never printed or embedded in the prompt. This is more than operational hygiene: in an agentic system, secrets management is part of model governance because the research artifact should remain shareable without carrying privileged credentials.

The notebook also limits concurrency through a bounded worker pool. A swarm is therefore not equated with unbounded parallelism. Concurrency is an engineering parameter constrained by rate limits, cost, latency, and failure handling.

A common evidence layer

Code Unit 2 downloads three years of adjusted daily prices through `yfinance`, applies missing-data thresholds, forward-fills surviving series, and computes daily returns. The agents themselves do not browse for prices. This creates a clean provenance boundary: factual market observations are obtained by conventional code, while language models operate only on the orchestrator’s derived packet.

This common-state design has an important experimental interpretation. If all agents observe the same evidence but produce different assessments, dispersion can be attributed primarily to mandates and model stochasticity rather than to different data vintages. It is an institutional analogue of holding the information set fixed while varying the decision rule.

Feature engineering

Code Unit 3 computes one-, three-, six-, and twelve-month trailing returns, annualized volatility, downside volatility, maximum drawdown, and a simple twelve-month return-to-volatility measure. Selected variables are standardized cross-sectionally. If x_{ik} denotes feature k for security i , the standardized feature is

$$z_{ik} = \frac{x_{ik} - \bar{x}_k}{s_k}. \quad (5)$$

Standardization is useful because the features otherwise live on different scales. More importantly, the features are intentionally interpretable. The notebook is an experiment in swarm architecture, not an attempt to conceal the signal inside a high-dimensional proprietary factor model.

The market packet also contains the correlation structure required by diversification specialists. Thus every agent sees a compact representation of both stand-alone security behavior and cross-security dependence.

The sequencing of these code units establishes a reproducible chain from raw observations to model-readable evidence. Environment configuration is separated from data retrieval, and data retrieval is separated from feature construction. That division makes failures easier to diagnose. A missing API key is an execution problem; an unavailable ticker is a data problem; a NaN produced by an insufficient lookback is a feature problem. Without such separation, an apparently unusual agent opinion could be caused by a silent upstream defect rather than genuine analytical heterogeneity.

The choice of adjusted daily prices is economically material. Adjustments seek to make price histories comparable across stock splits and distributions, while daily sampling supplies enough observations for volatility and covariance estimates over a three-year window. Even so, researchers must record the vendor, query date, adjustment conventions, trading calendar, and missing-value policy. Market data are revised, delisted names can disappear from convenient APIs, and forward filling can create artificial zero returns if used carelessly. The notebook’s filters are therefore part of the experiment, not mere data-cleaning boilerplate.

Feature horizons also embody assumptions. One-month return emphasizes recent information but may be noisy; twelve-month return captures a slower trend but may react late to regime changes. Downside volatility differentiates harmful dispersion from upside variation, while maximum drawdown records path-dependent loss severity that ordinary volatility can obscure. Risk-adjusted momentum combines reward and variability but remains backward looking. Presenting these features together permits agents to reason about conflicts among horizons and risk definitions. It does not imply that the selected variables span all economically relevant information.

Cross-sectional standardization changes the question from “Is this return high in absolute terms?” to “Is it high relative to the current universe?” That is appropriate for ranking securities at a common decision date, but it can conceal shifts in the overall market environment. If all securities have poor absolute momentum, the least negative name may still receive a favorable standardized value. Regime specialists are intended to challenge that interpretation, yet a production system would also include absolute indicators, market-level state variables, and explicit benchmark

context.

The common packet should remain compact enough to fit reliably within the model context and structured enough to discourage superficial narrative. Useful enhancements include feature definitions with units, observation dates, missingness flags, sector labels, and uncertainty estimates for noisy statistics. Each enhancement increases information but also raises token cost and the possibility that agents focus on irrelevant detail. Packet design is therefore an experimental variable. Researchers should test whether added fields improve stability or merely create new channels for correlated reasoning errors.

Most importantly, deterministic feature engineering creates a falsifiable boundary. If later analysis finds that the swarm performs poorly, the researcher can distinguish failure of the evidence set from failure of the agent interpretations or allocation rule. The system can be rerun with alternative features while holding the roster and optimizer fixed. That modularity is essential for cumulative research because it permits controlled comparisons rather than wholesale redesign after every disappointing result.

Code Unit 4: Fifty Agents as a Designed Research Institution

The swarm contains ten specialist families, each instantiated under five variants. This 10×5 construction is the principal source of designed heterogeneity.

Family	Financial perspective	Primary tension introduced
Momentum	Persistence across 3m, 6m, 12m returns	Strength versus fragility
Reversal	Overextension and mean reversion	Trend versus correction
Defensive	Low volatility, downside risk, drawdown	Return versus resilience
Drawdown	Capital preservation	Upside versus historical loss severity
Efficiency	Return per unit of risk	Raw return versus risk-adjusted return
Diversification	Portfolio contribution	Stand-alone attractiveness versus covariance value
Cross-sectional	Relative ranking across features	Single metric versus joint consistency
Regime	Stability if recent conditions change	Current fit versus regime robustness
Skeptic	Red-team falsification	Consensus versus failure modes
Balanced	Integrated judgment	Avoidance of one-factor dominance

The five variants further perturb each family toward robustness, dispersion, downside asymmetry, ranking stability, or falsification. This is crucial. Fifty repetitions of one prompt would create computational redundancy rather than an institution. The notebook instead tries to create a portfolio committee whose members are systematically asked to care about different dimensions of the same evidence.

One should nevertheless avoid a false independence interpretation. All fifty agents use the same base model. Their errors, priors, and linguistic habits can therefore be strongly correlated. The swarm is a designed ensemble of perspectives, not a sample of fifty independent forecasters. This distinction parallels ensemble learning: diversity among base learners can improve aggregate behavior, but common misspecification remains common [5].

The ten-family taxonomy can be read as a set of institutional objective functions. Momentum agents reward persistence; reversal agents search for overextension; defensive and drawdown agents emphasize different aspects of capital preservation; efficiency agents compare return with variability; and diversification agents evaluate marginal portfolio usefulness. Cross-sectional agents focus on relative ordering, regime agents question temporal stability, skeptics search for reasons the apparent signal may fail, and balanced agents integrate competing considerations. Because these mandates can point in opposite directions, the design creates productive tension rather than numerical repetition.

Five variants within each family serve a second purpose. They allow the experiment to observe variation inside a mandate as well as variation across mandates. If all momentum variants agree while the family sharply disagrees with defensive agents, the conflict is primarily institutional. If momentum variants themselves disperse widely, the evidence may be ambiguous even within a shared objective. Family-level and agent-level dispersion therefore answer different questions. This nested structure could later support hierarchical models that estimate family effects, variant effects, and residual model stochasticity.

Roster design must avoid decorative personas. A persona is useful only if it changes the decision rule in a testable way. Labels such as “bold investor” or “experienced analyst” may produce stylistic variation without informational diversity. By contrast, instructions to prioritize downside asymmetry, challenge horizon consistency, or assess correlation contribution identify observable analytical tasks. A rigorous prompt archive should specify these differences precisely enough that reviewers can predict where agents ought to disagree. If two mandates yield indistinguishable score distributions over many runs, one may be redundant.

The fixed base model creates both control and limitation. Holding the model constant isolates the effect of mandates, which is desirable for an architectural experiment. At the same time, shared training data and reasoning tendencies can generate common-mode failure that aggregation cannot diversify away. A unanimous swarm may simply be fifty expressions of one latent bias. Future experiments should cross agent mandate with model family, temperature, prompt formulation, and perhaps independent feature subsets. Such factorial designs would reveal whether apparent consensus survives genuine model diversity.

The number five per family is not asserted to be optimal. It balances coverage and execution cost while enabling within-family summaries. Marginal-value analysis could allocate more agents to families whose opinions are unstable or historically informative and fewer to consistently redundant families. An adaptive roster, however, introduces selection effects and should be evaluated using rules fixed before outcomes are observed.

Ultimately, the swarm is best viewed as a synthetic committee charter. Its value depends on whether membership produces distinct, relevant, and auditable perspectives. Headcount alone does not create collective intelligence; structured heterogeneity does. The taxonomy supplies hypotheses about where diversity should originate, while the opinion tape and convergence study provide the data needed to test whether those hypotheses are borne out.

Roster evaluation should therefore examine behavioral evidence, not persona labels. Distinct score distributions, rationale themes, sensitivity to features, and out-of-sample contributions can confirm whether a mandate has operational meaning. Agents that sound different but decide identically add theater rather than diversification.

Code Units 5–7: Contracts, Execution, and the Opinion Tape

Native structured output as an institutional contract

Code Unit 5 defines a JSON schema requiring `agent_id`, `family`, `portfolio_thesis`, `portfolio_risks`, and an array of ticker-level assessments. Each assessment contains a ticker, a score, confidence, and rationale. The score scale is bounded from -5 to $+5$ and confidence from zero to one.

The use of a schema is economically important. An investment institution cannot allow every research node to invent its own message format. Structured output turns a probabilistic language model into a component with a controlled interface. The content remains generative, but the contract is machine-verifiable. The current notebook also uses a larger output budget and explicitly rejects calls whose stop reason is not a normal end turn, addressing the truncation problem that arises when a long multi-security JSON object exhausts its token allowance.

The prompt further prohibits unsupported fundamentals, news, valuation data, forecasts, or events. This creates an epistemic constraint: the agent may interpret the supplied statistics but may not enlarge the evidence set by invention.

Parallel execution without deliberative herding

Code Unit 6 executes all fifty calls with bounded concurrency. Importantly, the agents do not communicate in this first-round architecture. Each sees the same market packet but not the opinions of peers. This resembles a pre-meeting investment committee in which analysts submit independent memos before discussion. The benefit is reduced herding. The cost is that agents cannot learn from a strong argument discovered by another specialist.

Future systems could add a second deliberation round, but that would change the object being measured. Independent first-pass opinions permit a cleaner interpretation of consensus and disagreement.

Validation and normalization

Code Unit 7 treats every model response as untrusted input. Parsed assessments are checked against the permitted ticker universe, scores are clipped to $[-5, 5]$, confidence is clipped to $[0, 1]$, and malformed responses are isolated. The resulting table has one row per agent–security pair:

$$\mathcal{O} = \{(j, i, s_{ji}, c_{ji}, r_{ji})\}, \quad (6)$$

where s_{ji} is agent j 's score for security i , c_{ji} its confidence, and r_{ji} its rationale. With fifty valid agents and thirty retained securities, the ideal tape contains 1,500 judgments.

This representation is analytically powerful because it converts a collection of natural-language interactions into a panel that can be studied with ordinary statistics. The system can ask which families are systematically optimistic, which securities generate disagreement, whether confidence is associated with extremity, and whether the same ranking survives when agents are removed.

The structured contract should be understood as a form of institutional law. It specifies not only the fields an agent must return but also the scope of permissible claims. A valid response identifies the agent, preserves the assigned family, covers recognized securities, uses bounded numerical scales, and supplies rationales tied to the provided

evidence. This reduces the distance between natural-language reasoning and downstream computation. Free-form prose may be persuasive to a human reader, but it is difficult to validate, aggregate, or compare consistently across fifty calls.

Schema compliance does not guarantee semantic validity. An agent can return syntactically correct JSON while offering a rationale that contradicts its score, omitting important evidence, or expressing unjustified confidence. Validation should therefore operate at several levels. Structural validation checks types and required fields. Domain validation checks tickers, ranges, duplicates, and coverage. Semantic diagnostics can flag contradictions, repeated boilerplate, implausibly uniform scores, or rationales that cite unavailable information. Human review remains valuable for sampled responses, especially when a new prompt or model version is introduced.

The opinion tape is the central audit artifact because it preserves information that a final portfolio would destroy. Aggregation compresses as many as 1,500 judgments into roughly thirty security-level statistics; optimization compresses those statistics further into a small vector of weights. Without the tape, a reviewer cannot reconstruct whether a position arose from broad moderate support, a narrow cluster of extreme scores, or an artifact of missing responses. Long-form rationales, family labels, timestamps, model metadata, and validation flags should therefore remain linked to every numerical observation.

Missingness deserves explicit treatment. If difficult securities generate more incomplete assessments, simply averaging available scores can introduce selection bias. The count of contributing agents should accompany every aggregate, and minimum-coverage rules should determine eligibility. Researchers can also compare the families of missing agents to detect systematic operational gaps. An unavailable skeptic is not equivalent to an unavailable momentum variant when the objective is balanced institutional judgment.

Normalization and clipping are safeguards, not substitutes for diagnosis. Clipping an out-of-range score prevents downstream failure, but frequent clipping may indicate that prompts or schemas are misunderstood. Production systems should track validation rates over time, establish alert thresholds, and quarantine runs that violate coverage expectations. In this sense, Code Units 5–7 create the equivalent of a controlled research intake process: agents may generate probabilistic judgments, but only well-formed, attributable, and reviewable observations enter the collective record.

Code Unit 8: Robust Collective Aggregation and the Economics of Disagreement

Code Unit 8 is the conceptual center of the notebook. It does not ask which agent is correct. It asks how a governed institution should summarize fifty bounded judgments without erasing disagreement.

For each security i , the notebook computes the median score $\tilde{s}_i$, mean score $\bar{s}_i$, standard deviation d_i , average confidence $\bar{c}_i$, positive share, and number of contributing agents. Collective conviction is defined as

$$C_i = \frac{\tilde{s}_i \bar{c}_i}{1 + d_i}. \quad (7)$$

Equation (7) is deliberately simple and inspectable. It has three economically intuitive properties. First, stronger median preference raises conviction. Second, higher stated confidence scales that preference. Third, disagreement reduces conviction even when the median is favorable.

The median is preferable to the mean as the primary location statistic when a small number of agents may produce extreme judgments. It has a 50% breakdown point in the classical robust-statistics sense and therefore resists a limited number of eccentric nodes [7]. The mean is still reported because the gap between mean and median can itself reveal skewness in the opinion distribution.

Disagreement should not be interpreted merely as noise. In asset pricing and investment management, dispersion of beliefs can reflect genuine uncertainty, heterogeneous models, or heterogeneous objectives. In the notebook, it is specifically a measure of model-and-mandate disagreement conditional on the same evidence. A high d_i says that the security's apparent attractiveness is institutionally contested.

Confidence deserves more caution. Language-model confidence is not a calibrated posterior probability. It is a self-reported scalar generated under the prompt contract. The notebook therefore uses average confidence as a modest multiplier rather than as a probability of correctness. A more advanced design would calibrate confidence empirically against subsequent ranking stability or realized outcomes.

Finally, the family matrix computes mean scores by ticker and specialist family. This is a governance device. An overall conviction number can conceal a conflict in which momentum and cross-sectional agents are enthusiastic while defensive and drawdown agents dissent. The matrix preserves the institutional topology of that disagreement.

Equation (7) should be interpreted as a policy rule rather than a discovered law. Its multiplicative numerator requires directional preference and stated confidence to reinforce one another, while its denominator imposes a smooth penalty

for dispersion. If the median score is zero, conviction is zero regardless of confidence. If agents agree strongly but report low confidence, conviction remains modest. If confidence and median score are high but opinions are widely dispersed, the rule recognizes that the central view is contested. These properties are transparent enough to debate and replace.

The use of standard deviation in the denominator is convenient but not uniquely robust. Median absolute deviation, interquartile range, entropy over score bins, or disagreement measured after grouping by family could produce different rankings. Standard deviation is sensitive to extreme opinions, which may be useful if an outlying specialist is signaling an important tail risk, but it can also over-penalize a single malformed judgment. Comparing alternative dispersion penalties is therefore a natural robustness test. The selected rule should be fixed before evaluating performance to avoid specification mining.

Confidence creates a separate research program. Model-generated confidence may reflect linguistic convention, prompt wording, or score extremity rather than empirical reliability. Calibration would require pairing archived forecasts with later outcomes or with reproducibility under resampling. Researchers could form confidence bins and examine whether higher-confidence judgments are more stable across reruns, more consistent with family mandates, or more predictive out of sample. Until such evidence exists, confidence should remain a bounded weight with limited influence rather than a literal probability.

Family aggregation can reveal minority information that the global statistic suppresses. Suppose eight families mildly favor a security while defensive and skeptic families strongly oppose it. The median may remain positive, yet the pattern suggests asymmetric downside concern rather than random noise. Governance dashboards should therefore display the global conviction beside family means, dispersion, coverage, and selected rationales. A portfolio manager can then determine whether a high score represents broad agreement or a coalition with identifiable dissent.

The aggregation layer also enables counterfactual analysis. One can remove a family, cap individual influence, equalize family weights, or recompute conviction under alternative confidence treatments. If rankings change dramatically when five closely related agents are omitted, the apparent consensus is fragile. If rankings remain stable across reasonable aggregation rules, confidence in the institutional signal increases. Robust collective judgment is thus not the output of one formula alone; it is the persistence of conclusions across defensible summaries of the same opinion tape.

WHY DISAGREEMENT BELONGS IN THE SIGNAL

A swarm that averages away dissent can become more confident precisely when it should become more cautious. Code Unit 8 instead treats dispersion as a denominator in collective conviction. The design is not claimed to be uniquely optimal; its virtue is that the institutional attitude toward disagreement is explicit, auditable, and replaceable.

Code Unit 9: From Collective Preference to a Constrained Portfolio

Code Unit 9 performs a sharp transfer of authority. The generative layer has ended. From this point forward, portfolio construction is deterministic.

The candidate set consists of up to twelve securities with positive swarm conviction. If fewer than five pass the positivity filter, the code falls back to the highest-ranked available names. Let E denote this eligible set. Conviction is min-max normalized:

$$\alpha_i = \frac{C_i - \min_{k \in E} C_k}{\max_{k \in E} C_k - \min_{k \in E} C_k + \varepsilon}, \quad i \in E. \quad (8)$$

The vector α is not an estimated expected return in the econometric sense. It is a normalized preference score. This distinction is essential. Calling it “alpha” is computational shorthand, not a claim that the swarm has estimated conditional expected excess returns.

The risk model is the annualized historical covariance matrix Σ calculated from daily returns. The optimization then solves Equation (3) using SLSQP. The long-only restriction removes leverage and short selling; the 15% cap limits single-name concentration; and full investment ensures all capital is allocated.

The resulting first-order intuition is straightforward. If constraints were absent and Σ were nonsingular, a quadratic program of this type would produce weights related to $\Sigma^{-1}\alpha$. In the actual notebook, inequality bounds and full investment make the solution piecewise constrained. Assets with attractive conviction can still receive less weight when they add substantial covariance risk, while a moderately attractive security may earn capital because it improves diversification.

This is where the notebook becomes a finance system rather than a voting exercise. The swarm produces preferences; portfolio theory prices the risk of expressing them jointly.

Why the optimizer must remain outside the swarm

One could ask each agent to propose weights and then average them. The notebook intentionally refuses that architecture. Weight generation is a constrained numerical problem with exact requirements. A language model can explain the constraints, but deterministic code can enforce them. This distinction is analogous to separating an analyst's recommendation from the portfolio-management system that verifies mandate limits.

The code also raises an exception if the optimizer fails. Silent fallback would undermine auditability. In a production system, one would additionally verify KKT residuals, constraint tolerances, numerical conditioning, and perhaps solve the same problem with an independent convex optimization package.

The transformation from conviction to α deserves careful interpretation because min-max normalization is relative to the selected candidate set. The highest-conviction eligible security receives a value near one and the lowest near zero, even if the absolute distance between them is economically small. A change in one extreme observation can rescale all candidates. Alternatives include rank scores, z-scores with winsorization, monotonic nonlinear mappings, or Bayesian shrinkage toward zero. Each method encodes a different belief about how ordinal agent preferences should translate into reward differences.

Candidate selection and optimization interact. Limiting the universe to twelve names reduces computational complexity and focuses the risk model, but it prevents excluded securities from contributing diversification. A name with only modest positive conviction could be valuable if it has low covariance with the favored group. Conversely, a highly ranked set may be crowded into correlated exposures. A useful extension would compare preselection with optimization over the full positive-conviction universe, perhaps using sparsity or transaction-cost penalties to control portfolio breadth.

Feasibility should be checked before the solver is called. With a 15% maximum weight, fewer than seven eligible securities cannot sum to one. The notebook's fallback rule seeks to maintain a workable candidate set, but a general implementation should calculate the minimum number of positions implied by the cap and verify all bounds explicitly. It should also distinguish infeasibility from numerical nonconvergence. These conditions have different remedies and should generate different audit messages.

The optimizer's output must be evaluated as a solution, not accepted as one because a success flag is present. Post-solution checks should confirm that weights sum to one within tolerance, remain inside bounds, correspond to the intended ticker order, and reproduce the reported objective value. Ex ante volatility, reward contribution, marginal risk contribution, concentration measures, and active constraints should be reported. Such diagnostics translate an opaque weight vector into an explainable allocation decision.

Sensitivity analysis is essential because optimized weights can be unstable. Researchers should perturb convictions, covariance entries, λ , candidate count, and the position cap. If small, plausible perturbations cause large reallocations, the portfolio is expressing estimation error rather than durable preference. Weight shrinkage toward equal weight, turnover penalties, or robust optimization can reduce instability. Comparison with a simple conviction-weighted or equal-weighted portfolio also provides a useful reality check: complexity should improve a defined objective sufficiently to justify its additional model risk.

This layer illustrates the architecture's central allocation principle. The agents do not choose weights, but their judgments still matter through a controlled channel. The optimizer does not decide which narratives are persuasive, but it determines how costly those preferences are when expressed together. Separating those roles makes both sides easier to audit and prevents fluent text from bypassing exact financial constraints.

Code Unit 10: Is Fifty an Informational Number or a Theatrical Number?

A large swarm should justify its scale empirically. Code Unit 10 therefore treats the number of agents as an experimental variable. It constructs the full-swarm mean-score ranking and repeatedly samples smaller groups of 5, 10, 20, 30, and 40 agents. For each sample it computes a ticker ranking and compares it with the full-swarm ranking using Spearman's rank correlation.

If R_i and Q_i are ranks under the full and subswarm systems, Spearman correlation is the Pearson correlation of the rank vectors. In the no-tie textbook case,

$$\rho_S = 1 - \frac{6 \sum_i (R_i - Q_i)^2}{n(n^2 - 1)}. \quad (9)$$

The notebook repeats the experiment 100 times at each swarm size and reports the mean, 10th percentile, and 90th percentile of ρ_S . This creates a convergence band rather than a single anecdotal comparison.

The test is conceptually important but should be interpreted correctly. The full fifty-agent ranking is used as the reference, not realized future returns. High correlation therefore means that a smaller swarm reproduces the full swarm's internal judgment; it does not establish predictive accuracy. The experiment measures marginal stability of

collective cognition.

This distinction suggests two separate research questions for future work. The first is internal convergence: how many agents are needed to stabilize the committee's ranking? The second is external validity: does a more stable ranking forecast anything economically useful out of sample? The current notebook answers only the first.

The convergence curve provides more information than a single average correlation. Its lower percentile indicates how poorly a randomly selected subswarm can perform relative to the full committee, while the upper percentile shows what is possible when the sample happens to be representative. A rising mean with narrowing bands suggests that additional agents reduce sampling uncertainty. A plateau suggests diminishing informational returns. Irregular behavior may indicate family imbalance, influential agents, ties in the rankings, or a small number of securities whose ordering is unstable.

Random sampling across all agents can overrepresent some families in small subswarms. This is informative if the operational question is how an arbitrary reduced committee behaves, but it does not isolate the value of roster size from the value of family coverage. A stratified audit could sample one or more agents per family, while a family-ablation audit could remove whole specialist groups. Comparing random and stratified curves would show whether stability comes primarily from headcount or from maintaining institutional diversity.

The full swarm is a convenient reference but not an independent truth. Because every subswarm contributes to the full-swarm ranking, correlations are mechanically related, and larger samples overlap more heavily with the benchmark. A stronger internal study could use repeated split halves, bootstrap consensus rankings, or a held-out panel of agents generated under separate seeds or models. These alternatives would not establish economic value, but they would provide a less self-referential measure of reproducibility.

Rank correlation also ignores the economic size of score differences. Swapping two nearly tied securities has the same rank effect as swapping two names with widely separated convictions. Complementary diagnostics could measure score-vector correlation, top- k overlap, weighted rank correlation, or portfolio-weight distance after optimization. If the practical output is a portfolio, stability of selected holdings, turnover, and risk exposures may matter more than stability of the complete ranking.

Cost belongs in the scale analysis. Each additional agent consumes tokens, latency, retry capacity, and validation effort. The economically relevant question is not whether fifty agents are more stable than five, but whether the improvement from agent m to agent $m + 1$ is worth its marginal resource cost and operational complexity. A scale frontier can plot convergence quality against monetary cost and wall-clock time. Different applications may choose different points: a classroom demonstration, overnight research process, and intraday system face very different constraints.

Finally, scale should be linked to outcomes only under a prospective protocol. At each historical decision date, a frozen swarm would form a ranking using information then available; subsequent returns and risks would be recorded; and performance would be compared across swarm sizes using identical portfolio rules. Such an experiment could determine whether internal stability is beneficial, irrelevant, or even harmful. A committee can converge reliably on a shared mistake. Code Unit 10 is therefore a diagnostic of institutional consistency, not a substitute for market validation.

Reporting should include the random seeds and exact sampled agent identifiers for every repetition. This permits unusually strong or weak subswarms to be reconstructed and inspected, turning the convergence chart from a summary graphic into an auditable experiment.

Statistical Interpretation: The Swarm as an Ensemble of Conditional Decision Rules

The agent system can be understood without anthropomorphism. Each agent is a conditional mapping

$$A_j : \mathcal{X} \times M_j \rightarrow \mathcal{S}, \quad (10)$$

where $\mathcal{X}$ is the common market packet, M_j the mandate and variant, and $\mathcal{S}$ the structured score space. The swarm is the collection $\{A_j\}_{j=1}^{50}$ followed by an aggregation functional G .

This perspective connects the notebook to ensemble methods. Bagging reduces variance by averaging models trained on perturbed samples; boosting sequentially focuses on errors; random forests decorrelate trees through feature and sample randomness. The notebook uses a different source of diversity: semantic role perturbation over a shared foundation model. Its diversity is therefore neither classical sampling diversity nor independent model diversity. It is *mandate diversity*.

That difference has consequences. If the base model contains a systematic misconception about a financial concept, all fifty agents may inherit it. More agents will not necessarily diversify common bias. Conversely, if the model has several plausible internal reasoning pathways, mandate diversity may cause different pathways to be expressed,

allowing robust aggregation to exploit useful variation. The empirical question is whether role-induced diversity is informative enough to justify its cost.

The notebook’s skeptic family is especially interesting in this interpretation. It is not an independent adversarial model; it is a prompt-conditioned attempt to draw from a critical mode of the same model. A stronger research design would compare same-model role diversity with multi-model diversity and with human analyst diversity.

Formally, agent j may be viewed as implementing an unknown stochastic rule f_j that maps the common state $\mathcal{X}$, mandate M_j , prompt specification P_j , and generation randomness ϵ_j into a score vector. The observed output is one realization of

$$s_j = f_j(\mathcal{X}, M_j, P_j, \epsilon_j). \quad (11)$$

This representation makes clear that cross-agent variation combines designed mandate effects with uncontrolled stochastic effects. Repeating the same agent under several seeds or independent calls would help estimate the latter. Without reruns, a difference between two variants cannot be attributed exclusively to their mandates.

Ensemble gains usually depend on error diversity. If individual errors are unbiased and imperfectly correlated, aggregation can reduce variance. Language-model agents violate the simplest independence assumptions because they share architecture, training data, prompts, and evidence. Their error covariance may be high, particularly on questions that trigger a common learned heuristic. The central empirical object is therefore not the number of agents but the dependence structure of their errors. That structure can only be estimated against repeated tasks with observable outcomes or credible labels.

The current aggregation rule is robust in the cross-section of agents at one date, but statistical inference requires a time dimension. With repeated decision dates, researchers could estimate the persistence of family effects, the stability of score distributions, and the relation between dispersion and future volatility or forecast error. Mixed-effects models could separate security, date, family, and agent components. Such analysis would show whether named specializations correspond to stable behavioral differences rather than prompt-level rhetoric.

Conditional decision rules also invite regime analysis. A momentum family may add value in persistent markets and fail around reversals; defensive agents may appear overly cautious during calm periods but informative before stress. Instead of assigning fixed global weights, a meta-model could learn family reliability conditional on observable states. This must be done cautiously and out of sample, because adaptive weighting adds another layer vulnerable to overfitting. Simple equal family influence remains a strong benchmark.

Bias and variance have institutional analogues here. Adding similar agents may reduce idiosyncratic generation variance while leaving common bias unchanged. Adding genuinely different mandates may reduce some forms of bias but increase dispersion. Penalizing all disagreement mechanically can then suppress useful minority signals. Researchers should distinguish noise-like disagreement from structured dissent associated with specialist expertise. Family matrices, rationales, and later outcomes provide evidence for that distinction.

The statistical interpretation ultimately reframes the swarm as a data-generating process that can itself be studied. Prompts, mandates, outputs, failures, and revisions become observations. This perspective discourages anthropomorphic claims about a committee “thinking” and replaces them with testable questions about stability, dependence, calibration, and incremental predictive content. It is the appropriate foundation for deciding whether a multi-agent architecture offers more than repeated sampling from one model.

Accordingly, claims about ensemble benefit should be accompanied by uncertainty estimates across dates, seeds, and roster compositions. A single successful run is an anecdote; a stable distribution of improvements under a frozen design is evidence.

Financial Interpretation: Conviction, Uncertainty, and Risk Are Different Objects

A recurring mistake in AI-finance prototypes is to collapse all uncertainty into one score. The notebook instead contains at least three distinct objects.

First, *conviction* is a collective preference about relative attractiveness. Second, *disagreement* is cross-agent dispersion in that preference. Third, *market risk* is the covariance structure of realized security returns. These quantities are related but not interchangeable.

A stock can have high conviction and low disagreement but still be highly volatile. Conversely, it can have moderate conviction, substantial disagreement, and strong diversification value. The optimization layer makes the latter distinction visible because Σ affects weights independently of C_i .

This decomposition is analogous to institutional investment practice. Research confidence is not the same as portfolio risk; analyst disagreement is not the same as volatility; and a compelling thesis does not mechanically imply a large position. Position sizing is where beliefs encounter the joint distribution of returns and mandate constraints.

Conviction is directional and conditional. It summarizes how the committee interprets the supplied evidence relative to the current universe. It is not a probability that a security will rise, an estimate of expected excess return, or a guarantee that the underlying thesis is complete. A positive conviction can be economically useful only if the evidence set contains relevant information and the mapping from features to judgments has predictive content. The notebook therefore treats conviction as an input to allocation rather than as a sufficient statistic for investment value.

Disagreement is epistemic rather than market based. It measures variation among model-and-mandate judgments given a common packet. High disagreement may indicate ambiguous features, conflict among objectives, sensitivity to prompt framing, or useful specialist dissent. Low disagreement may reflect genuinely clear evidence, but it may also reveal common-model bias. The interpretation depends on how disagreement behaves across reruns, families, models, and realized outcomes. A governance process should examine its structure rather than assuming that less is always better.

Market risk concerns the distribution of returns after a position is taken. Historical variance and covariance are estimates of that risk, but they do not measure uncertainty about the swarm's reasoning. Two securities can receive identical conviction and disagreement statistics while contributing very different amounts of portfolio volatility. Conversely, a volatile security may improve portfolio efficiency when its covariance with other holdings is low. The optimizer preserves this distinction by applying Σ after the research signal has been formed.

Estimation uncertainty constitutes a fourth object. Conviction, disagreement, and covariance are all estimated or constructed from finite information. The sample covariance matrix may change with the lookback window; collective conviction may change with roster composition or generation randomness; and confidence may change with wording. Sensitivity analysis should therefore report ranges rather than presenting a single portfolio as uniquely determined. Robustness across plausible specifications is evidence of stability, not proof of truth.

These distinctions improve communication with decision makers. A portfolio report can state that a security has strong committee support, identify whether the support is broad, disclose its standalone and marginal risk, and show how stable the proposed weight is under perturbation. Collapsing these statements into one number would obscure their different meanings. Clear terminology also prevents confidence theater, in which a fluent system appears precise because several incomparable uncertainties have been compressed into a score.

The practical lesson is that belief formation and risk bearing answer different questions. Research asks what appears attractive and why. Aggregation asks how much the research institution agrees. Risk modeling asks what losses and co-movements are plausible. Portfolio construction asks how strongly the institution should express its beliefs under constraints. The notebook's layered design is valuable precisely because it refuses to answer all four questions with the same model output.

This separation also clarifies how new information should enter the system. Evidence that changes the expected attractiveness of a security belongs in the research packet and may alter conviction. Evidence that changes expected co-movement or loss severity belongs in the risk model and may alter sizing even if conviction is unchanged. Evidence about mandate limits belongs in the constraint set. Mixing these channels makes attribution difficult and encourages double counting.

A complete decision report should therefore present at least four panels: the directional conviction and its components; the distribution of family views; standalone and marginal portfolio risk; and sensitivity of the proposed weight to alternative assumptions. Decision makers can then see whether a large position is driven by unusually strong belief, diversification value, loose constraints, or a fragile numerical estimate.

The taxonomy is useful beyond this notebook. In credit, one might separate default conviction, model disagreement, spread volatility, and portfolio concentration. In macro allocation, one might distinguish regime belief, scenario dispersion, asset covariance, and risk budget. The general principle is unchanged: uncertainty about a judgment should not be represented by the same variable used to measure the market consequence of acting on it.

Governance: Autonomy by Layer, Not Autonomy Everywhere

The notebook offers a useful definition of governed autonomy. Agents are autonomous within a narrow analytical mandate: they may interpret, rank, explain, and criticize. They are not autonomous with respect to data acquisition, ticker eligibility, schema, score bounds, portfolio constraints, or numerical optimization.

This layered design resembles the principle of least privilege in computer security. A component should receive only the authority required for its function. In finance, the analogue is segregation of duties. The research desk may propose; the risk system may constrain; the portfolio manager may allocate; compliance may prohibit. Combining all rights in one opaque agent increases operational and model risk.

The structured-output contract also creates provenance. Every judgment retains agent identity and family. A final position can therefore be traced backward from weight to conviction, from conviction to distribution of scores, and from scores to specialist rationales. This is far more defensible than a single chat response that says "buy these stocks."

Layered autonomy can be represented as an authority matrix. The data service may retrieve and clean observations but may not form opinions. Agents may interpret approved evidence but may not expand the ticker universe or place trades. Validation may admit or reject responses but may not alter their economic meaning silently. Aggregation may summarize accepted judgments but may not bypass minimum coverage rules. The optimizer may assign weights within the mandate but may not override prohibited securities or concentration limits. Every capability is paired with an explicit prohibition.

This design limits the blast radius of failure. If an agent hallucinates an unsupported event, validation or review can isolate one response. If the optimizer receives an ill-conditioned matrix, it can halt without corrupting the opinion archive. If market data are stale, the entire run can be invalidated before any new positions are authorized. Monolithic autonomous systems make such containment difficult because data access, reasoning, sizing, and execution occur inside one opaque trajectory.

Governance also requires versioned change control. Prompt revisions, schema changes, model upgrades, feature additions, covariance methods, and constraint changes can all alter outcomes. Each should receive a version identifier, rationale, test record, and effective date. Comparative runs should be conducted before deployment, with special attention to validation rates, score distributions, concentration, and turnover. A hosted model name alone is insufficient because providers may update behavior behind a stable interface.

Human oversight should be designed rather than invoked vaguely. Reviewers need dashboards that show exceptions, missing coverage, family conflicts, data freshness, active constraints, and deviations from prior runs. Intervention rights should be specified: who may reject a run, change an eligible universe, approve a model upgrade, or authorize execution? Overrides should be logged with reasons so that later performance attribution does not mistakenly credit or blame the swarm for a human decision.

Security and operational controls remain relevant even in a research notebook. API credentials should be isolated, prompts should not expose sensitive holdings, external text should be treated as potentially adversarial, and logs should avoid unnecessary confidential information. Rate limits and service outages require deterministic failure behavior. If the model is unavailable, the correct response may be to preserve the prior approved portfolio or halt, not to improvise with partial output.

Governed autonomy is therefore not reduced autonomy. It is purposeful delegation within enforceable boundaries. Agents receive enough freedom to contribute heterogeneous interpretation, while capital-moving authority stays with transparent mechanisms and accountable people. The architecture's credibility comes from this asymmetry: creativity is distributed widely, but permission to alter the portfolio is narrow, explicit, and auditable.

Regular governance review should test whether these boundaries still match the system's actual behavior. New tools, retrieval sources, memory, or model capabilities can silently expand what an agent is able to do. Periodic access reviews, red-team exercises, and end-to-end simulations should verify that documented permissions remain effective. Governance is therefore a continuing operational process, not a diagram prepared once at launch.

Accountability must also survive vendor dependence. An institution cannot outsource responsibility by pointing to a model provider. It remains responsible for choosing the service, validating outputs, setting limits, monitoring changes, and deciding whether results influence capital. Contracts and technical controls may allocate tasks, but fiduciary and managerial responsibility remains human and institutional.

WHAT THE NOTEBOOK DOES NOT ESTABLISH

The notebook does not establish that Claude Sonnet 5 forecasts equity returns, that fifty agents outperform one agent, that the portfolio has positive expected alpha, or that the historical covariance matrix is an adequate production risk model. It is an architectural and pedagogical experiment. Its claims concern decomposition, governance, interpretability, and internal convergence unless a separate out-of-sample return experiment is added.

Relationship to Established Portfolio Methods

Markowitz and constrained mean–variance optimization

The final allocation is recognizably mean–variance in spirit [9]. The important departure is the source of the reward vector. Rather than estimating sample means, the notebook maps collective AI judgment into a normalized score. This can reduce the temptation to interpret noisy historical average returns as precise forecasts, but it does not make the new signal statistically valid by itself.

Black–Litterman as a natural extension

The Black–Litterman framework is especially relevant because it was designed to combine an equilibrium prior with subjective views [2]. A future version could treat swarm outputs as views $P\mu = q$ with uncertainty matrix Ω .

Cross-agent disagreement could help parameterize Ω : high disagreement would imply less confidence in a view. This would give Code Unit 8 a direct Bayesian portfolio interpretation rather than the current heuristic normalization.

Covariance shrinkage and robust optimization

Historical sample covariance is unstable in high dimensions, though the current eligible universe is deliberately small. Ledoit–Wolf shrinkage [8] or factor covariance could replace the raw estimator. Robust optimization could also recognize uncertainty in both α and Σ rather than optimizing as though they were known [3].

Naive diversification as a benchmark

DeMiguel, Garlappi, and Uppal show how difficult it can be for optimized portfolios to outperform simple $1/N$ allocation out of sample once estimation error is considered [4]. The notebook should therefore be extended with an equal-weight benchmark, minimum-variance portfolio, risk-parity portfolio, and perhaps a conventional momentum strategy. Without those comparisons, one cannot attribute performance value to the swarm.

The notebook can also be situated within the broader distinction between signal combination and portfolio combination. In signal combination, the fifty judgments are aggregated before one optimizer forms a portfolio. In portfolio combination, each family or agent could produce a separately constrained portfolio and those portfolios could then be blended. The first approach exposes one common risk model and encourages direct comparison of opinions. The second permits specialist objectives to survive farther downstream but makes aggregate exposures harder to control. Testing both would reveal whether information is lost by early aggregation.

Black–Litterman provides a particularly disciplined route for translating subjective judgments. A market-implied prior could anchor expected returns, while the swarm supplies relative or absolute views. The view matrix P would specify which securities or spreads are involved, q would encode view magnitude, and Ω would represent uncertainty. Dispersion, coverage, historical reliability, and confidence calibration could all inform Ω . This framework would prevent normalized conviction from being interpreted as a standalone expected-return estimate and would make uncertainty enter the posterior explicitly.

Robust portfolio methods address a different weakness: optimization acts as though estimated inputs were known. Uncertainty sets around rewards and covariance can penalize allocations that depend on fragile estimates. Resampling, shrinkage, and regularization offer related protections. The swarm’s disagreement measure could contribute to security-specific uncertainty bounds, but this link should be justified empirically. High agent dispersion does not automatically imply high return-estimation error, even though that is a plausible hypothesis.

Factor models provide another natural benchmark and diagnostic. Final weights can be decomposed into market, size, value, momentum, quality, sector, and other exposures. If the swarm portfolio is largely a conventional momentum portfolio expressed through elaborate prompts, the factor analysis will reveal that. This does not make the system useless, but it changes the attribution of novelty. Incremental performance should be evaluated after controlling for transparent factor exposures.

Simple portfolios remain essential controls. Equal weight tests whether optimization adds value at all. Minimum variance isolates the contribution of the covariance model. A feature-only momentum portfolio tests whether agents improve on the numerical evidence they receive. A single-agent portfolio tests the value of institutional diversity. A random-score portfolio helps identify benefits produced mechanically by constraints. Comparing these nested benchmarks allows researchers to assign gains or losses to the research signal, aggregation, and allocation layers separately.

Established methods therefore do not compete with the swarm as obsolete alternatives. They supply priors, risk estimators, optimization tools, benchmarks, and skepticism about estimation error. The most credible path is integration: use generative agents where contextual judgment might add information, and use classical finance to determine whether that information survives disciplined comparison and constrained expression.

Performance attribution should mirror this integration. Return differences can be decomposed into universe selection, agent-derived security ranking, aggregation, covariance-aware sizing, constraints, and trading effects. Without such decomposition, a favorable portfolio result may be credited to the swarm even when it arose from a conventional factor tilt or risk-model choice. Layer-specific attribution converts a broad architectural claim into testable component claims.

This comparison also protects against novelty bias. A method should earn adoption by improving a relevant outcome relative to simple, transparent alternatives. If it does not, the notebook still retains pedagogical value, but its role in investment production should remain limited.

A Formal Research Agenda for Testing Economic Value

The current notebook is an architecture demonstration. A PhD-level empirical extension should introduce a strict temporal design. At date t , features would be computed only from information available through t ; the swarm would score securities; the optimizer would form weights; and returns over $[t, t + h]$ would be recorded. The process would roll forward without look-ahead.

The central hypotheses could be written as follows. Let R_{t+1}^S be the swarm portfolio return, R_{t+1}^B a benchmark return, and C_t a measure of average conviction.

$$H_{0,1} : E[R_{t+1}^S - R_{t+1}^B] = 0, \quad (12)$$

$$H_{0,2} : \text{RankCorr}(C_{i,t}, R_{i,t+1}) = 0, \quad (13)$$

$$H_{0,3} : \text{Performance}(N = 50) - \text{Performance}(N = k) = 0, \quad (14)$$

for smaller swarm sizes k .

A serious study would account for transaction costs, turnover, survivorship bias, delistings, corporate actions, multiple testing, and the fact that repeated LLM calls may change as model versions change. Standard errors would need to reflect time-series dependence. If many alternative prompt architectures were tried, the research process itself would create specification-search risk analogous to backtest overfitting [1].

The economic evaluation should include annualized return, volatility, Sharpe ratio, maximum drawdown, turnover, information ratio relative to a benchmark, factor exposures, tail loss, and implementation shortfall. Predictive rank correlation should be reported separately from portfolio performance because a weak but stable ranking can still have value under careful sizing, while a strong in-sample ranking can fail after costs.

The empirical protocol should be specified before results are observed. Researchers must choose rebalance frequency, holding horizon, data lag, universe rules, transaction-cost model, benchmark, and treatment of failed agent calls. Prompts and model identifiers should be frozen for each experimental phase. Any modification made after examining performance creates a new specification and should be evaluated on a fresh period. This separation between development, validation, and final test samples is difficult but necessary when prompt iteration is inexpensive.

A point-in-time universe is essential. Selecting today's surviving liquid equities and projecting them backward introduces survivorship and selection bias. Each rebalance date should use securities that were genuinely eligible then, with delistings and corporate actions represented. Feature computation must respect publication and availability lags. If later extensions add fundamentals or macroeconomic data, revisions and reporting delays must also be controlled. The same discipline applies to text: news published after the decision timestamp cannot enter the evidence packet.

The study should use multiple evaluation levels. Security-level tests can examine rank information coefficients, calibration of positive and negative scores, and whether disagreement predicts absolute forecast error. Portfolio-level tests can measure returns, risk, drawdowns, costs, turnover, capacity, and factor exposures. System-level tests can measure latency, failure rates, token cost, and reproducibility. A method that appears predictive but fails operationally is not deployable; a reliable system with no incremental economic information is not an investment strategy.

Inference must account for overlapping returns, autocorrelation, heteroskedasticity, and repeated model selection. Block bootstrap or appropriate time-series standard errors can improve uncertainty estimates. Performance should be reported across subperiods and regimes rather than through one full-sample statistic. Confidence intervals and effect sizes are more informative than isolated significance thresholds. If many variants are compared, multiple-testing adjustments or a locked final holdout reduce the risk of discovering a strategy by chance.

Ablation studies are especially important for architectural claims. Remove confidence from the conviction rule, remove the dispersion penalty, collapse the roster to one family, replace the model with deterministic feature rankings, or bypass optimization with equal weights. Each ablation tests whether a component contributes incremental value. Interaction tests can determine whether, for example, disagreement helps only when covariance risk is high or whether specialists add value only in certain regimes.

Finally, the research record should include unfavorable runs and abandoned specifications. Selective reporting is especially tempting when model calls generate many plausible variants. A preregistered or time-stamped experiment ledger can document hypotheses, changes, and stopping rules. The objective is not to produce the best historical chart; it is to estimate whether governed multi-agent judgment adds repeatable economic information after realistic costs and uncertainty.

Replication by an independent team would provide an especially strong test. The replicators should receive the frozen protocol and archived inputs but implement the pipeline separately. Agreement would demonstrate that results do not depend on hidden notebook state or undocumented judgment. Disagreement would reveal which supposedly mechanical choices require clearer specification.

The Economics of Swarm Scale

Fifty agents impose direct token cost and indirect latency. Let $K(N)$ denote computational cost for a swarm of size N and $I(N)$ the informational value of its aggregate ranking. Rational scale requires considering marginal value:

$$\Delta V(N) = I(N + 1) - I(N) - [K(N + 1) - K(N)]. \quad (15)$$

The notebook's convergence experiment provides an internal proxy for the first term: if rank stability saturates quickly, additional calls may have low marginal value.

This suggests adaptive swarm sizing. Rather than always launching fifty agents, an orchestrator could begin with ten strategically diverse agents, estimate ranking stability and disagreement, and expand only when uncertainty remains high. Securities with high disagreement could receive additional specialist attention while obvious cases terminate early. Such an architecture would turn swarm size into an endogenous computational budget.

The broader lesson is that “more agents” is not a scientific objective. The objective is sufficient diversity and redundancy to improve decision quality at acceptable cost.

Informational value must be defined before it can be optimized. It may refer to higher out-of-sample rank correlation, lower variance across repeated runs, better detection of tail risks, more stable portfolio weights, or broader coverage of plausible arguments. These objectives are not equivalent. A larger swarm might stabilize the average score while adding little predictive value, or it might surface rare but important objections without changing the central ranking. Scale decisions should be tied to the institution's actual loss function.

The shape of $I(N)$ is likely to depend on roster composition. Adding a sixth near-duplicate momentum agent may contribute less than adding the first liquidity or valuation specialist. Marginal value should therefore be measured by both headcount and perspective. Shapley-style contribution analysis, leave-one-family-out tests, or sequential additions under several orderings can estimate which agents improve stability or performance. Because contributions may interact, no single ordering is definitive, but repeated designs can identify persistent redundancy.

Operational cost extends beyond API charges. Larger swarms require more validation, storage, monitoring, retry handling, and human review. They also increase the probability that at least one response is malformed or delayed. Tail latency can dominate completion time when the system waits for every agent. A practical orchestrator may use deadlines, minimum coverage thresholds, or asynchronous acceptance rules. Those choices alter the composition of the final committee and therefore belong in the research design.

Adaptive stopping offers a principled alternative to a fixed roster. After an initial stratified group responds, the orchestrator can estimate whether rankings, dispersion, and family coverage have reached predefined tolerances. Additional agents are launched only if the result remains unstable. To avoid optional-stopping bias, thresholds must be specified in advance and evaluated on held-out periods. The audit record should show the path by which the swarm expanded, including which uncertainty triggered each additional call.

Allocation of computation can also vary by security. Most names may generate consistent judgments, while a few sit near the eligibility threshold or display intense family conflict. Targeted follow-up agents could analyze only those ambiguous cases. This resembles active learning: computational effort is directed where it is expected to reduce decision uncertainty. However, selective attention can change score comparability, so follow-up results should be marked separately from the first-round tape and aggregated under explicit rules.

Scale may have negative returns. Larger groups can amplify common framing, create false confidence through repetition, and make human oversight harder. If the aggregation rule treats each correlated variant as independent evidence, five nearly identical agents can dominate one distinctive specialist. Family weighting and correlation-aware aggregation can mitigate this problem. The effective number of independent perspectives may be far smaller than the nominal number of calls.

The economic question is consequently one of system design under scarcity. Tokens, time, review capacity, and risk budget are limited resources. A rational swarm uses enough agents to achieve a measured level of diversity and stability, then stops. Fifty is a useful experimental scale, not a universal optimum.

A deployment decision can formalize this trade-off through a service-level objective. For example, the system might require a minimum family coverage, maximum expected latency, bounded cost per rebalance, and a target lower-percentile ranking stability. Candidate swarm configurations can then be compared on a common frontier rather than selected by intuition.

Scale choices should be revisited as model prices, latency, universe size, and evidence complexity change. A roster that is efficient for thirty securities and one daily run may be inefficient for hundreds of securities or intraday updates. The optimum is conditional on the task and infrastructure.

Failure Modes and Model Risk

Several failure modes deserve explicit attention.

Common-model bias. All agents share Claude Sonnet 5, so role diversity cannot eliminate model-wide misconceptions. Multi-model replication would provide a stronger robustness test.

Prompt-induced correlation. The mandates are distinct but use a common vocabulary and evidence packet. Similar prompt structure may induce correlated scoring behavior.

Confidence miscalibration. Self-reported confidence may reflect linguistic style rather than empirical reliability. It should be calibrated or downweighted if it adds no predictive information.

Historical-state insufficiency. The market packet is price-based. It contains no valuation, earnings, balance-sheet, macroeconomic, options-implied, or textual information. Agents are correctly forbidden from pretending otherwise, but the investment signal is consequently narrow.

Covariance instability. Historical covariance can change sharply across regimes. A three-year estimate can understate future stress correlations.

Objective-function arbitrariness. The choice $\lambda = 4$, the 15% cap, the top-12 eligibility rule, and the conviction formula are design parameters. They require sensitivity analysis.

Reference dependence in the audit. Code Unit 10 compares subswarms with the full swarm, not with economic truth. Convergence can measure consistency without measuring accuracy.

Data-provider limitations. The demonstration uses `yfinance`; a production or publishable empirical study should use institutional-grade data with explicit corporate-action and survivorship controls.

Schema-valid semantic failure. A response can satisfy every required field while misreading a feature, contradicting itself, or repeating generic language. Automated checks should compare stated rationales with numerical inputs and sample outputs for human review.

Missing-not-at-random responses. Agents may fail more often on complex prompts or ambiguous securities. Aggregates based only on successful responses can then appear more certain than the complete committee would have been. Coverage thresholds and failure-pattern analysis are necessary.

Roster imbalance. Equal weighting at the agent level gives greater influence to any perspective represented by more variants. Even with five agents per family, prompts may overlap conceptually. Family-balanced aggregation and leave-family-out tests can reveal hidden concentration of influence.

Regime and lookback mismatch. Three years of daily data may combine distinct market environments or omit an earlier stress episode. Features and covariance can respond differently to the chosen window. Multiple lookbacks and stress scenarios should accompany the base estimate.

Optimization instability. Small changes in conviction or covariance can move constrained weights sharply, especially near active bounds. Sensitivity analysis, regularization, and comparison with simpler allocations should be mandatory before interpreting exact weights.

Prompt and model drift. Hosted language models can change even when an interface name remains stable. Prompt edits intended to improve formatting may also alter economic judgments. Archived raw outputs, version metadata, and benchmark test packets are required to detect drift.

Automation bias. Reviewers may defer to a polished dashboard or a fifty-agent consensus without inspecting evidence quality. Interfaces should foreground uncertainty, dissent, failures, and assumptions rather than presenting the portfolio as a singular recommendation.

Security and data leakage. Future systems that ingest proprietary holdings, research, or client constraints could expose sensitive information through prompts or logs. Access control, minimization, redaction, and vendor governance are part of model risk.

Execution mismatch. A theoretically feasible portfolio may be impractical after spreads, market impact, taxes, lot sizes, restricted lists, or trading windows. Portfolio construction should incorporate implementation constraints before any live use.

These risks are not independent. Stale data can interact with confident common-model bias; missing skeptic responses can interact with roster imbalance; and unstable optimization can magnify a small semantic error. Risk assessment should therefore include end-to-end scenarios rather than isolated component tests. For example, a stress exercise might combine a volatility regime shift, several timed-out agents, and a binding concentration cap to observe whether the system halts safely or produces a misleadingly precise allocation.

Controls should be classified as preventive, detective, and corrective. Schemas, evidence boundaries, and position limits prevent some failures. Coverage dashboards, drift tests, sensitivity analysis, and reconciliation detect others. Retry policies, rollback procedures, model disablement, and human overrides provide correction. Each control needs an owner, threshold, and recorded outcome. A warning that nobody is responsible for reviewing is not an effective control.

Model risk also includes the possibility that the system works exactly as designed but the design has no economic value. Clean code, valid JSON, stable rankings, and feasible weights can coexist with zero predictive content. Benchmarking

and out-of-sample testing remain the decisive safeguards against mistaking operational sophistication for investment skill.

Residual risk should be accepted explicitly. No control can guarantee that the data are complete, the model is unbiased, or future covariance resembles history. Governance should state which residual risks are tolerated, by whom, and at what capital scale. Unresolved uncertainty is a reason to limit exposure and increase monitoring, not to conceal it behind additional automation.

Pedagogical Interpretation for Graduate Finance

The notebook is particularly useful because it forces several concepts that are often taught separately to interact in one system.

First, it makes clear that a financial signal and a portfolio are different mathematical objects. The swarm produces a cross-sectional preference; covariance and constraints determine how that preference can be expressed in capital.

Second, it makes disagreement observable. In a conventional single-model pipeline, uncertainty is often compressed into a residual variance or confidence interval. Here the institution itself produces a distribution of views.

Third, it demonstrates that generative AI does not eliminate the need for deterministic code. On the contrary, the more generative the upstream component becomes, the more valuable explicit contracts, validators, numerical solvers, and audit trails become downstream.

Fourth, it provides a concrete setting for discussing model risk. Students can change one design element at a time—the number of families, the aggregation rule, the risk-aversion parameter, the maximum weight, the covariance estimator—and observe how institutional architecture changes outcomes.

Fifth, it shows why “agent” should be understood functionally. An agent is not intelligent because it has a name. It becomes useful when it has a mandate, evidence boundary, output contract, provenance, and a defined place in a larger decision process.

The notebook can support a sequence of classroom experiments. Students can first reproduce the deterministic market packet and verify every feature by hand for a small subset of securities. They can then inspect how different mandates transform identical evidence into different scores. Next they can construct the opinion tape, calculate medians and dispersion, and test alternative aggregation rules. Finally, they can solve the constrained portfolio problem and explain why rankings and weights differ. This progression links data engineering, behavioral interpretation, statistics, and optimization.

Assignments should emphasize hypotheses rather than cosmetic prompt changes. A student might predict that removing defensive agents will increase exposure to high-volatility names, or that family-balanced aggregation will differ from agent-balanced aggregation when one family is expanded. The notebook then becomes a laboratory in which an institutional change has a measurable consequence. Requiring predictions before execution helps prevent retrospective storytelling.

The architecture is also useful for teaching the limits of inference. High subswarm correlation does not prove return predictability. A successful solver does not prove that its inputs are accurate. A persuasive rationale does not prove causal understanding. A backtest with many prompt variants may overfit just as readily as a conventional quantitative strategy. Students can learn to separate computational correctness, internal validity, external validity, and economic significance.

Several quantitative exercises follow naturally. Learners can derive the unconstrained solution to a simplified mean-variance objective, identify when the maximum-weight constraint binds, compare sample and shrinkage covariance, or calculate marginal risk contributions. They can simulate score distributions to see how medians, means, and dispersion penalties behave under outliers. They can also study how a fixed number of correlated agents produces less diversification than independent votes.

Qualitative analysis remains equally important. Students can code rationales by theme, examine whether agents comply with evidence restrictions, and evaluate whether skeptic arguments identify genuine failure modes. Comparing the narrative record with numerical scores can reveal contradictions and boilerplate. Such work demonstrates that structured output makes language analyzable without pretending that semantic quality is fully captured by a scalar.

The notebook supports discussion of professional ethics and accountability. If a model-generated portfolio harms a client, responsibility cannot be assigned to an anthropomorphized swarm. Designers choose the data, mandates, aggregation, constraints, and deployment process; reviewers approve changes; institutions decide whether outputs influence capital. Clear authority mapping therefore belongs in finance education alongside mathematical technique.

Assessment can culminate in a reproducible research memo. Students should state an intervention, archive the relevant prompts and data, report validation failures, compare benchmarks, and distinguish observed results from interpretations. The goal is not to reward the most favorable portfolio. It is to reward disciplined reasoning about what the experiment can and cannot show.

In this way, the swarm serves as a compact model of a modern investment organization. It makes visible the often-hidden transitions from evidence to opinion, opinion to consensus, consensus to risk-adjusted weights, and weights to governance. That visibility is its primary pedagogical strength.

Because every transition is represented in code and data, students can audit the full chain rather than studying isolated formulas. They learn that professional rigor comes from the connections among methods: clean evidence, bounded judgment, robust statistics, appropriate optimization, honest validation, and accountable decision rights.

THE NOTEBOOK IN ONE SENTENCE

Common market evidence is interpreted by fifty deliberately heterogeneous agents; their structured judgments are validated and aggregated into a disagreement-aware collective preference; deterministic portfolio theory then converts that preference into constrained weights; and a final audit asks whether the scale of the swarm actually stabilizes the ranking.

Extensions Toward an Advanced Autonomous Investment System

The current architecture is intentionally one-shot. It can be extended along several dimensions without sacrificing governance.

A first extension is temporal memory. Agents could compare the current state with prior committee states and identify which theses are persisting, weakening, or reversing. Memory should be structured rather than conversational: previous scores, realized outcomes, and forecast errors can be stored in tables.

A second extension is outcome-based reputation. If agent j repeatedly contributes useful rank information out of sample, its future vote could receive greater weight. Formally, one could estimate reliability parameter ω_j and aggregate

$$C_i^{(w)} = G(\{\omega_j s_{ji}, c_{ji}\}_{j=1}^N). \quad (16)$$

Care would be required to avoid overfitting short histories or creating self-reinforcing dominance.

A third extension is deliberation. After an independent first round, a small critic panel could receive anonymized arguments and identify contradictions. A second round could permit agents to revise scores. The research question would then become whether deliberation improves out-of-sample performance or merely increases herding.

A fourth extension is dynamic constellation formation. Instead of fixed ten-family architecture, a supervisor could inspect the market state and allocate more agents to relevant problems: volatility specialists during stress, reversal specialists after extreme dispersion, or liquidity specialists when turnover constraints bind. This would connect the notebook to adaptive multi-agent systems rather than fixed committees.

A fifth extension is explicit human governance. A portfolio manager could approve the candidate universe, adjust mandate constraints, or veto positions for information unavailable to the notebook. Human intervention should itself be logged so that performance attribution distinguishes swarm, optimizer, and discretionary overlays.

A sixth extension is retrieval from governed internal sources. Agents could receive earnings transcripts, filings, research notes, or macroeconomic releases through a retrieval layer that records document identifiers and timestamps. Retrieval must remain separate from free browsing so that every factual claim can be traced to an approved source. Conflicting documents and stale information should be surfaced rather than silently reconciled by the model.

A seventh extension is scenario generation. Specialist agents could propose coherent shocks to rates, growth, volatility, or sector relationships, while deterministic code translates those scenarios into portfolio losses using factor or full-revaluation models. The generative contribution would be the breadth and narrative coherence of scenarios; numerical risk calculation would remain controlled. Scenario plausibility and duplication would need review.

An eighth extension is explicit transaction and liquidity modeling. Expected spreads, market impact, turnover, tax considerations, and capacity can be added to the optimization objective. This would force swarm preferences to overcome the real cost of changing positions. It would also reduce the temptation to interpret frequent opinion changes as actionable information.

An advanced system could maintain a model registry containing each agent's mandate, prompt version, base model, reliability history, allowed tools, and retirement status. New agents would enter through a shadow period in which their outputs are recorded but do not affect capital. Promotion would require evidence of distinctiveness, operational reliability, and incremental value. Existing agents could be retired when redundant or persistently unhelpful.

Deliberation requires special caution. Showing agents a consensus can induce anchoring, while exposing identities can create artificial deference patterns learned from language. Anonymized arguments, independent scoring before discussion, limited revision rounds, and explicit preservation of minority reports can reduce herding. The system should record both pre- and post-deliberation views so that the effect of discussion is measurable.

Autonomy could eventually extend to experiment proposal rather than trade execution. Agents might identify

unstable assumptions, suggest ablations, or request additional data. A deterministic scheduler and human review board would decide which experiments run. This arrangement uses generative systems to expand the research agenda while maintaining control over resources and production changes.

Live deployment would require a staged path: historical replay, paper trading, shadow production, limited capital, and only then broader use. Each stage needs predefined performance, risk, reliability, and rollback criteria. A system that passes predictive tests but fails data freshness or audit completeness should not advance. Conversely, operational reliability cannot substitute for economic evidence.

The guiding principle across all extensions is that added capability must be paired with added observability and control. Memory requires versioned state; reputation requires out-of-sample calibration; retrieval requires provenance; deliberation requires comparison with independent views; dynamic routing requires logged selection rules; and human intervention requires attribution. Advanced autonomy is credible only when the institution can reconstruct why the system acted and which authority permitted it.

Reproducibility and the Status of the Companion Notebook

The companion notebook is not merely illustrative code; it is the executable specification of the architecture discussed here. The paper therefore makes explicit reference to its ten code units. The mapping is summarized below.

Unit	Function	Research role
1	Environment and secrets	Reproducibility, model identity, credential separation
2	Market evidence	Common factual state and data provenance
3	Feature engineering	Interpretable cross-sectional state and covariance inputs
4	50-agent roster	Designed heterogeneity: 10 families × 5 variants
5	Structured contract	Bounded epistemic authority and machine-readable output
6	Concurrent execution	Independent first-pass collective cognition
7	Validation	Convert probabilistic outputs into an auditable opinion tape
8	Aggregation	Robust location, disagreement, confidence, family map
9	Portfolio construction	Deterministic constrained optimization
10	Swarm audit	Internal convergence and marginal scale diagnostics

Reproducibility in an LLM system differs from reproducibility in purely deterministic code. The same prompt can yield different language-model outputs, and hosted model versions can evolve. A rigorous research archive should therefore save the notebook, exact prompts, model identifier, SDK version, execution timestamp, raw structured responses, parsed opinion tape, market-data snapshot, optimization parameters, and final portfolio. Where possible, hashes should identify immutable artifacts.

Three levels of reproducibility should be distinguished. Computational reproducibility asks whether the archived code and inputs regenerate the deterministic features, aggregates, and optimization results. Procedural reproducibility asks whether another researcher can execute the same sequence of model calls under the same contract and validation rules. Statistical reproducibility asks whether repeated runs produce substantively similar rankings and portfolios despite stochastic generation. The first can often be exact; the second depends on access to services and model versions; the third must be reported as a distribution rather than a binary property.

The market-data snapshot is especially important. A ticker list and date range are insufficient because vendors can revise adjusted histories, correct errors, or change missing observations. The archive should include the actual price matrix used, metadata describing adjustments, and a hash of the file. Feature code should record lookback conventions, annualization factors, missing-data thresholds, and the decision timestamp. This allows a reviewer to distinguish a data revision from a change in agent behavior.

Model calls require an equally detailed manifest. Each record should contain the stable agent identifier, family, variant, complete system and user prompts, schema version, model identifier, generation settings, request time, response time, stop reason, token usage, retry history, raw response, and validation status. Secrets must never be archived with the research artifact. If the provider exposes a model snapshot identifier, it should be retained; if not, benchmark prompts can help detect later behavioral drift.

Randomness should be controlled where possible and characterized where it cannot be controlled. Numerical sampling in the subswarm audit can use stored seeds, making those results exactly repeatable. Hosted generation may not honor a reproducible seed or may still change across infrastructure versions. The appropriate response is to save outputs and run repeated-call robustness studies. Reproducibility does not require pretending that a stochastic service is deterministic; it requires documenting the stochasticity and preserving the realized research record.

The notebook itself should be executable from a clean environment with pinned package versions. Cells should run in order without hidden state, and expensive external calls should be separable from deterministic downstream

analysis. A useful archive stores raw responses so that aggregation, visualization, and optimization can be rerun without incurring new model calls. Environment files, platform information, and dependency versions help explain discrepancies caused by changing libraries or solver behavior.

Artifacts should form a lineage graph. The final portfolio points to an optimizer configuration and covariance matrix; the reward vector points to a conviction table; conviction points to the validated opinion tape; the tape points to raw responses and prompt versions; and the prompts point to the exact market packet. Stable run identifiers can connect these objects. This lineage makes it possible to audit a single weight backward without relying on narrative memory.

Reproducibility also benefits from automated tests. Unit tests can verify feature calculations and score clipping. Contract tests can submit known responses to the validator. Integration tests can run a small mock roster through aggregation and optimization. Regression tests can compare a fixed packet with archived summary statistics while tolerating documented stochastic variation. Before publication or deployment, a clean rerun should confirm that all tables and reported results derive from the archived artifacts.

Finally, the status of the notebook should be stated precisely. It is an executable companion and architectural prototype dated September 2026, not a frozen production trading system. Its evidentiary claims remain bounded by the data and experiments it contains. Preserving that distinction in the archive prevents later readers from treating pedagogical parameter choices or illustrative outputs as validated investment recommendations.

Conclusion

The 50-agent financial swarm is best understood not as an attempt to replace portfolio management with a crowd of language models, but as an experiment in institutional design. It asks what happens when generative intelligence is given a legitimate but bounded role inside a quantitative process.

The answer is a layered architecture. Conventional code establishes a common market state from historical prices. Fifty Claude Sonnet 5 agents, organized into ten specialist families and five variants, interpret that state through deliberately different financial mandates. Native structured output transforms their judgments into a common contract. Validation converts those contracts into a panel of agent–security opinions. Robust aggregation then recognizes that collective intelligence is not simply an average: the median captures central preference, confidence scales the strength of that preference, and disagreement reduces conviction. Family-level matrices preserve the institutional structure of dissent.

At that point the language model’s authority ends. The portfolio layer selects eligible securities, normalizes conviction, imports the historical covariance matrix, and solves an explicit constrained quadratic objective. Long-only positions, full investment, and a maximum 15% position are enforced by numerical optimization rather than by natural-language instruction. This separation is one of the notebook’s most important lessons. Generative systems can contribute judgment without being granted authority over every stage of a financial decision.

The final audit introduces an equally important discipline: scale must be tested rather than admired. By comparing smaller random subswarms with the full fifty-agent ranking, the notebook asks whether additional agents stabilize the collective ordering. The test does not prove economic value, because the full swarm is an internal reference rather than future realized returns. It does, however, establish the right research instinct. A large swarm should earn its computational complexity through measurable marginal information.

For finance, the deeper implication is that agentic AI can be integrated without abandoning the intellectual machinery developed over decades of portfolio theory. Markowitz diversification remains relevant. Covariance remains relevant. Constraints remain relevant. Estimation error remains relevant. So do model risk, provenance, and institutional separation of duties. What changes is the research layer: instead of one fixed signal model, we can create a population of bounded analytical perspectives and study the distribution of their judgments.

The next scientific step is therefore not to make the swarm more theatrical. It is to make the experiment harder. Freeze the data vintages. Roll the system through time. Compare against equal weight, minimum variance, momentum, factor models, and Black–Litterman variants. Include transaction costs and turnover. Test multiple base models and prompt architectures. Calibrate confidence. Measure whether disagreement predicts future uncertainty. Estimate the marginal value of the twentieth, thirtieth, and fiftieth agent. Only then can one move from architectural plausibility to evidence about investment performance.

The notebook nevertheless establishes a powerful pedagogical principle. A financial AI system becomes more credible when intelligence is distributed but authority is not. The agents may disagree, interpret, criticize, and propose. Deterministic mathematics still governs admissibility and capital allocation. The resulting system is neither a conventional factor model nor an unconstrained autonomous trader. It is a hybrid institution in which collective generative judgment and quantitative finance occupy different, explicit, and auditable roles.

The architecture therefore offers a practical standard for future systems. Every generative contribution should have a defined evidence boundary, every output should enter through a validated interface, every aggregate should preserve enough detail to reveal dissent, and every capital decision should satisfy explicit mathematical and institutional

constraints. Scale should be justified by measured marginal value, not by the appearance of sophistication. When these conditions are absent, a swarm can multiply opacity faster than it multiplies insight.

The decisive empirical question remains open. A governed system can be coherent, reproducible, and internally stable without forecasting returns or improving portfolios after costs. The appropriate response is not to weaken governance but to subject the architecture to prospective, time-stamped, benchmarked research. Such testing can identify whether mandate diversity contributes unique information, whether disagreement is a useful uncertainty signal, and whether any benefits survive risk adjustment and implementation.

Even before that evidence exists, the notebook demonstrates a durable design lesson. Generative AI need not displace classical finance to become useful within it. Its role can be limited to producing diverse, inspectable interpretations of approved evidence, while conventional statistics and optimization retain authority over measurement and allocation. This division of labor makes the system easier to challenge, improve, and, when necessary, stop.

The objective of a financial swarm is not to replace discipline with more intelligence. It is to place more intelligence inside a disciplined architecture.

References

[1] Bailey, D. H., Borwein, J. M., López de Prado, M., and Zhu, Q. J. (2014). Pseudo-mathematics and financial charlatanism: The effects of backtest overfitting on out-of-sample performance. *Notices of the AMS*, 61(5), 458–471.

[2] Black, F., and Litterman, R. (1992). Global portfolio optimization. *Financial Analysts Journal*, 48(5), 28–43.

[3] Boyd, S., and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press.

[4] DeMiguel, V., Garlappi, L., and Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies*, 22(5), 1915–1953.

[5] Dietterich, T. G. (2000). Ensemble methods in machine learning. In *Multiple Classifier Systems*, Lecture Notes in Computer Science 1857, 1–15.

[6] Fama, E. F., and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.

[7] Huber, P. J. (1981). *Robust Statistics*. Wiley.

[8] Ledoit, O., and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.

[9] Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.

[10] Michaud, R. O. (1989). The Markowitz optimization enigma: Is optimized optimal? *Financial Analysts Journal*, 45(1), 31–42.

[11] Reynoso, A. (2026). *A 50-Agent Financial Swarm: From Distributed Judgment to a Governed Equity Portfolio*. Executable Jupyter/Google Colab research notebook: `NB_50_AGENT_FINANCIAL_SWARM_CLAUDE_SONNET_5.ipynb`.

[12] Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.

[13] Wooldridge, M. (2009). *An Introduction to MultiAgent Systems* (2nd ed.). John Wiley & Sons.

[14] Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. *International Conference on Learning Representations (ICLR)*.

[15] Wu, Q., et al. (2023). AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*.

[16] Hong, S., et al. (2024). MetaGPT: Meta programming for a multi-agent collaborative framework. *International Conference on Learning Representations (ICLR)*.

COMPANION COMPUTATIONAL ARTIFACT

The executable research record for this paper is `NB_50_AGENT_FINANCIAL_SWARM_CLAUDE_SONNET_5.ipynb`. It contains the market-data layer, engineered state, 50-agent taxonomy, Claude Sonnet 5 structured-output contract, bounded concurrent execution, validation and opinion tape, disagreement-aware aggregation, deterministic constrained portfolio construction, and the subswarm convergence audit described in this paper.